

Computational Supplement

Accepted Service Adaptation: Continuation Prices, Transfer Coordinates, and Certified Gains

Revision R24, September 23, 2026

CS.1 R24 diagnostic protocol and complete numerical records

This computational supplement preserves every earlier empirical table and execution detail relocated from the electronic companion. It also supplies the new interior-comparator and repeated-query diagnostics. It is a code-and-data supplement, not a substitute for a proof required to read the main paper. All predecessor formal statements and proofs remain in the main manuscript or electronic companion. Sources and records retain their original revision labels; old data are not pooled into new sampling claims.

CS.1.1 Interior-model comparator brackets

The coefficient seed is 2402301. For every original context and radius, two three-coordinate integer vectors are sampled uniformly from minus seven through seven and divided by eight. Every vector is strictly interior to the cube. Both comparators use the true interior reward and friction. The true restricted class uses that model’s own participation and capacity coefficients; the outer class is exactly the original analytically containing class. Both retain root, box, and time-only amendment restrictions.

Each solver supplies a proposal and dual prices. A separate rational procedure rounds date/service amplitudes, reconstructs root payments, and mixes within the time-only subspace toward a strict accepted anchor. A general rational Fenchel calculation then checks the two comparator intervals without trusting solver status or its printed objective. Negative controls perturb root payment, time-only equality, inequality prices, switching tensions, and interpolation weights. The records include each rational implementation and all dual prices. The mean, median, quartiles, ninety-fifth percentile, and maximum of total, outer-set, and interpolation slack are retained at every radius. This is a sensitivity diagnostic, not a statistically calibrated uncertainty model.

CS.1.2 Timing, training, and memory accounting

Training seeds are 2402401 through 2402403; test seeds are 2403401 through 2403403. Each configuration has one fixed fit and a fresh held-out sample. The predictors retain the earlier 192-feature tanh-gradient construction and cubic radial-basis direct-price construction. Audited optimizer-generated labels contain both the optimized value and its resource statistic; they are not information learned without an optimizer. The entire label generation, including the teacher workspace and rational audit, is charged to either learned method. Shared labels are not charged twice when the two methods are jointly fitted, but a standalone deployment of either method carries the full label bill.

A setup timer includes sparse matrix construction and workspace initialization. Online timers include every candidate-specific operation, including numerical repair and exact checking. Method order is independently randomized within a context. The classical comparator retains both primal and dual continuation

state along the same ordered context sequence; each new repetition resets that sequence. No method is allowed to use another method’s fallback state. When fallback occurs it starts in a new lifted workspace. Consequently the cost record directly distinguishes proposal overhead from fallback avoidance and from hypothetical warm-start savings not implemented here.

A context is the resampling unit for paired intervals. Its three timings are averaged first, and the 32 context-level paired savings are then resampled 10,000 times with seed 2405401. These exploratory percentile intervals concern fixed fitted policies on the declared context distribution. Repetition does not turn 32 contexts into 96 independent training/deployment experiments. The retrospective R22 scaling reanalysis has only eight context-level pairs per configuration and is labeled accordingly.

Memory statistics distinguish serialized predictor bytes, reachable NumPy array storage, and sparse input-matrix bytes. Array views may share backing storage; sparse matrix counts exclude factorization workspace. Peak resident memory is a whole-process high-water mark, not a per-method peak. These quantities are reported with their scopes rather than summed into an unsupported total-memory comparison.

CS.1.3 Recorded robust certification cost

The original robust run stores each method’s optimization, rational repair, and vertex-evaluation time and the shared eight-comparator batch time. The retrospective analysis adds that entire batch to each method when interpreting it as an independently certified decision. It also reports the batch separately so that actual reuse across candidate methods is visible. Original ensemble prediction occurred before those timers; no new-machine measurement is spliced into the archived totals. The resulting archived sum is a disclosed lower bound on full learned end-to-end time, not evidence for acceleration. A pointwise gain certificate is an operationally more expensive object than a feasible proposal or a certificate used only in offline evaluation.

The fresh robust-cost check uses seed 2407401, sixteen contexts, and radii zero and one quarter. Ensemble prediction is timed on each query rather than precomputed outside the timer. Each of the eight comparator solves uses a fresh workspace and an independent rational upper certificate. Candidate ordering rotates by context. Model deserialization and geometry construction are one-time costs and are reported separately. These measurements are made sequentially after the nominal matched study, on the same execution host; they are not added to the historical R23 machine times. All rational records are independently replayed.

CS.2 R22 recorded objects and independent verification

Each timed record contains the initial repaired proposal and its dual certificate, the fixed-primal polished certificate, and the actually returned post-fallback certificate. Replay verifies identical proposal coordinates, switching tensions and inequality prices before and after polishing; nonincreasing rational upper bounds; and the requested final gap. Component checks evaluate the exact four-term identity, including its outward-rounding remainder. An outside gate, when active, is recomputed from the exact proposal value; an ungated component identity is not relabeled as the selected outside decision’s gap.

Validation records additionally retain the optimized restricted candidate and its full restricted dual, both deterministic ensemble outputs, reference decisions, and every corner bound. The sample observations

use the selected value minus the restricted upper bound. Both the Hoeffding target and its width are independently reconstructed from these objects. The complete dual-repair checks independently optimize 225 small accepted problems, including 45 with no strict-feasibility point; they verify 3,600 exact projected-gradient steps and 450 reward/friction certificate transports. They do not call the numerical optimizer or the main certificate implementation. These are theorem checks, not speed comparisons. The 360 nonsmooth zero-curvature identities and 234 exact regularization-bias checks supplement, rather than replace, the proofs. Historical rational replay remains executable without any numerical solver. Source hashes and final-manuscript hashes are verified on the published commit, rather than only on its pre-publication workflow input.

Reading the preserved R19 study. The following complete section is retained from R19, with its numerical values unchanged. References there to “new” or “current” computations describe R19, not R22. For its uniform-fleet theorem, the reported experiment uses one common context distribution for every constituent policy; a deployed policy index is drawn independently of that common context. More generally, policy-specific laws define the different joint mixture that first draws the index and then draws a context conditionally on it. The latter is not a guarantee for an index chosen independently of a common deployment context. R22’s deterministic ensemble validation avoids this target ambiguity altogether.

CS.3 Repeated learning, difficult geometry, and complete deployment cost

The empirical design tests the link in Section 5 rather than using the star’s easy scalar solve as evidence for difficult stochastic control. We retain the entire earlier information hierarchy, verified-cell studies, star experiments, and adverse nonlinear-transfer results in the companion and the immutable predecessor documents. The current multistage observations were produced in the R16 research run and are integrated into the manuscript here for the first time. Our fleet inference is a retrospective reanalysis of those observations. The accuracy-targeted cost study and deliberate boundary checks are new follow-up computations, separately identified in their execution records.

CS.3.1 A multistage, vector, nonsmooth study

The fixed primitive has 63 nodes on a binary tree with six review levels, two service coordinates per node, six coupled quadratic resources, and four additional cross-service capacity constraints. Outside tiers vary by node and service. Every nonroot subtree retains its payment cap, and each service has its root payment equality. Absolute switching costs include installation and have endogenous signs. The displacement formulation is exactly the quadratic instance following Theorem 5.2; it is not a collection of independent stars. Each context changes six resource-reward coordinates and a strictly positive friction coefficient. The complete rational primitive and context grids are recorded in the companion.

We evaluate eight independently seeded training/deployment pairs. Each pair uses 256 training contexts and 256 fresh deployment contexts. No best seed or model is selected. All seven methods use the same exact-solver label population: tanh random-feature scalar critics fitted with value-only or value-and-gradient

losses; a direct-price tanh regressor; the corresponding three quadratic response surfaces; and a cubic radial-basis direct-price interpolant. The hidden tanh weights are fixed random features, not a fully trained deep network. Value-only and derivative-supervised critics both deploy through their derivative. Direct-price methods predict the same six-dimensional operational statistic. Thus the comparison does not grant neural methods a different response operator or a more favorable certificate.

The feasible outside protocol is specified and node-dependent, not proved to be the reoptimized best static protocol for this heterogeneous primitive. This study tests accepted implementation relative to that announced comparator. The separate canonical and centered-friction studies retain their reoptimized static comparators. Monetary values in the current study are total normalized tree-objective units; they are not per-review gains from the earlier star normalization. The primitive is synthetic and not calibrated to a service provider. Those distinctions prevent a gain against one known feasible protocol from being advertised as an empirical universal value of adaptation.

The R16 solver amendment increased the iteration ceiling after a failure in the R15 execution, while retaining seeds, methods, loss weights, and tolerances; the full amended study was rerun. We describe this as a frozen, independently deployed internal experiment, not an externally preregistered study. Neither this reanalysis nor the follow-up computations modify its trained models or replace its unfavorable observations.

CS.3.2 Policy quality and the meaning of derivative supervision

Table 13 reports all seven methods on all 2,048 deployment contexts. The reference objective is bracketed by its primal value and independent dual upper bound. Therefore the reported regret is an upper bound, not an unverified subtraction from a numerical optimizer. The average reference bracket width is 8.11×10^{-8} .

The mean regret upper bound falls from 0.0343743 for value-only tanh fitting to 0.00368462 with derivative supervision. Across eight independent training/deployment pairs, the approximate paired Student interval for their mean difference is $[-0.0415827, -0.0197968]$. This comparison resolves the direction of improvement for these two fitted procedures in this distribution; it does not overwrite the statistically unresolved R14 star comparison. The independent unit is a complete training/deployment pair, not an individual test context or timing repetition.

Direct-price tanh fitting improves further, to 0.00158094. Quadratic direct-price and radial-basis fitting attain 0.000271149 and 0.000157737, respectively. The latter two also have much higher immediate certificate pass rates at 10^{-3} . Thus the derivative statistic is operationally useful while neural parameterization is not necessary in this study. This is a result of the required comparison, not a reason to remove neural methods, their mathematical guarantee, or their adverse observations. It identifies the stronger current implementation of the same accepted-control architecture.

CS.3.3 Conditional gain for an explicitly deployed frozen fleet

A simultaneous lower bound for each of 56 individual frozen policies, using only its own 256 contexts and the original analytical gain range of 25, is zero. We retain that result. Pooling 2,048 contexts as though they came from one selected model would be invalid. A different, explicitly implementable policy has a more

informative guarantee: for each new independent context choose one of the eight frozen policies uniformly, independently of that context, then execute its audited response and outside gate.

The conditional frozen-fleet proposition and its full proof remain in Electronic Companion Section EC.5. For $K = 8$, $n = 256$, $M = 7$, $B = 25$, and $\delta = 0.05$, the penalty is 0.868352. The resulting seven lower bounds, included in Table 13, are positive. They certify this finite, frozen randomization rule under the stated context distribution. The eight-pair Student comparison addresses training/deployment variability approximately; the fleet inequality instead gives a finite-sample, conditional guarantee. Neither substitutes for a guarantee over arbitrary retraining, misspecification, or distribution shift.

CS.3.4 Geometry rather than successful-region averages

The earlier R16 structural suite had no binding extra capacity among its 64 tested instances, despite some tier boundaries. We do not relabel those instances as a capacity stress test. The new 64-instance suite deliberately includes heterogeneous controls, root cross-service capacities with positive incentives, tier-boundary incentives with slack extra capacities, and high-friction switching kinks. Each stratum uses four primitive seeds and four contexts per seed on 15- or 31-node, two-service trees with two or four resource factors. All 16 cases in each targeted stratum must actually exhibit the specified geometry or the test fails. No failing instance may be dropped.

Every reference and repaired response is independently replayed in rational arithmetic. Theorem 5.2 is checked with its certified response error rather than by treating the numerical subproblem solution as exact. An independent SLSQP solve starts from the outside protocol on a capacity-binding instance. Finite-difference checks perturb only the resource-reward coordinate, keeping acceptance, switching, and all other primitives fixed. Exact rational scalar checks additionally exercise a quartic smooth resource coupling with nonsmooth switching. These are theorem tests, not new trained-policy observations or evidence of calibrated operations performance.

The retained star reassessment also includes all original, critical-friction, capacity-activation, and primitive-perturbation strata. Its capacity-activation stratum binds the root constraint in every case. All methods then require capacity response adjustment and yield the same certified solution to numerical precision. That is evidence for the response solver, not successful neural extrapolation. The companion reports stratification by friction distance, resource binding, leaf activation and saturation, nearest breakpoint, and optimum gain, without selecting only favorable cells.

CS.3.5 Accuracy-targeted total cost and amortization

The original R16 timings charge prediction, the base response, rational repair and audit, actual full-objective refinement, and a second audit. They use a 10^{-9} numerical tolerance even for the coarse deployment target. That design may over-solve the classical comparator. We preserve those measurements and execute the previously recorded, but unimplemented, accuracy-targeted comparison instead of treating the earlier apparent speed advantage as decisive.

In the follow-up, both classical solvers and all surrogate response solvers first use numerical tolerance

10^{-5} . Every method must pass the same independent full-objective certificate at the requested tolerance; otherwise it performs a full 10^{-9} refinement and another audit. The cached classical baselines use cold and previous-context starts. All seven frozen surrogate models are included. We use the first 16 contexts of every deployment cohort, two repetitions, and rotated method order. The total comparison contains 4,608 completed pipelines, with 256 records per method and requested tolerance. Timing observations and model-selection units are kept separate.

At the coarse target, the lower-mean classical baseline costs 13.453 ms, compared with 9.563 ms for radial-basis direct-price deployment. Its observed full-pipeline saving has a conservative offline break-even count of 2,546 queries. At the strict target, the corresponding classical cost is 26.180 ms and the radial-basis cost is 37.808 ms. These are separately measured follow-up results, not the original fixed-solver-tolerance timings.

The table reports mean complete cost, not the sum of separately reported medians. The companion supplies medians, 95th percentiles, fallback rates, and paired cohort intervals. Offline cost is measured on the same host by replaying all 256 training-label computations and all seven fits for every cohort, while leaving the deployed frozen models unchanged. We conservatively charge that entire shared offline cost to each surrogate. A finite break-even count is reported only when its observed mean online saving is positive. It is a benchmark-specific accounting calculation, not a future runtime guarantee. The data do not establish universal neural superiority, and no failed cost comparison is hidden behind a warm-start or an unaudited prediction time.

CS.4 Current study: primitives, audits, and reproducibility

CS.4.1 Primitive construction and the scope of the experiment

The 63-node binary tree has two service coordinates at every node. An unconditional coefficient for service a at node j is $c_{ja} = t_{ja}2^{5-\text{depth}(j)}$, with integer $t_{ja} \in \{8, \dots, 12\}$. Outside tiers have denominator 32 and numerators from 13 through 19. Diagonal curvatures have numerators 8 through 24 and denominator 16; marginal reward offsets have numerators -8 through 8 and denominator 32. The six-by-126 resource map has integer entries -4 through 4 divided by 32. Four nonnegative cross-service capacity rows have integer coefficients 0 through 4 and right-hand sides with denominator 8. All initial installed tiers are one half. These values are generated once using primitive seed 15001; the resulting exact arrays, not a verbal generator alone, are supplied.

Every nonroot subtree has the outside protocol’s payment cap separately for each service, while each service’s root payment is fixed. The displacement vector therefore obeys homogeneous continuation inequalities and two root equalities. The outside protocol is heterogeneous and is not asserted to solve a static optimization problem for this primitive. Context reward numerators are independent integers from -16 through 16 divided by 16, and the positive friction numerator is from 2 through 32 divided by 64. The coefficient of each absolute switching difference has a numerator from 4 through 8 and denominator 8. Primitive heterogeneity does not itself constitute field calibration.

The derivative in Theorem 5.1 uses the six reward coordinates, not the scaled friction coordinate. Training

rescales friction to a centered unit interval; all derivative targets are resource vectors in the reward coordinates’ original units. The 192 tanh random features include a constant and the seven linear input coordinates. The ridge parameter is 10^{-7} . Quadratic models use the corresponding linear and quadratic monomials; radial-basis direct-price interpolation uses a cubic kernel and smoothing 10^{-7} . All fitted coefficients and feature weights are serialized. The full generator, solver adapter, independent audit, and model definitions are retained in the R16 source directory.

CS.4.2 Feasibility and finite precision

For each service and nonroot node, negate its weighted subtree displacement and divide by its positive payment coefficient to obtain a transfer coordinate. Negative numerical transfers are truncated to zero and the rest are rounded downward on a grid with denominator 10^8 . The sparse two-entry-per-column transfer map reconstructs a rational displacement satisfying every root equality and continuation cap exactly. A common nonnegative radial contraction toward zero enforces all box and extra capacity rows. This construction works for the study’s positive payment coefficients and homogeneous outside-relative commitments; it is not claimed for arbitrary signed tariffs. The general theorem permits any independently certified feasible response instead.

Tensions are clipped to their exact friction capacities; inequality prices are clipped to nonnegativity. Prices and multipliers use denominator 10^9 . The independent verifier recomputes box feasibility, subtree sums, root equalities, resource capacities, tension capacities, exact primal gain, and an outward-rounded conjugate upper bound. It uses only Python integers and rational arithmetic, with no numerical solver or matrix inverse. In particular it does not trust the proposal’s active set, the repair routine’s assertions, or the solver’s termination flag. The gate compares exact gain with the known outside protocol only.

CS.4.3 Complete timing and offline accounting

Each target tolerance has its own previous-context classical workspace. This prevents a second target tolerance from obtaining an uncharged within-context warm start. Cold solvers reset their iterate; all solver matrices and factorizations that the solver caches are already constructed before repeated decisions. Initial numerical tolerance is 10^{-5} for both full classical and price-response proposals. Actual full-objective refinement uses 10^{-9} , initialized from the rejected proposal, with a new rational audit afterward. Prediction, response, repair, both audits when needed, and refinement are charged. The numerical tolerance is not the deployment certificate tolerance.

The first label in each cohort is cold; the subsequent offline labels use previous-solution warm starts. All 256 labels and all seven model fits are measured on the same host as the follow-up online run. Offline coefficients are not substituted for the already frozen deployment models. For a method with mean complete cost t_m and the lower observed classical mean t_c , the descriptive break-even count is the ceiling of charged offline cost divided by $t_c - t_m$, only when $t_m < t_c$. The comparison with the lower-mean classical method is descriptive: associated Student intervals are not multiplicity-adjusted claims after comparator selection.

CS.4.4 Hard-regime stratification and retained adverse evidence

The star reassessment has four training seeds, four explicit strata, 128 contexts per stratum and seed, and eight methods, yielding 16,384 method-context records. Methods share contexts within a seed. These are not 16,384 independent training replications. Stratum-level and geometry-level summaries are generated from every record; no stratum, zero-gain context, or failed immediate certification is discarded. The full machine-readable stratification also conditions on absolute distance to critical friction, active and saturated leaf counts, nearest breakpoint distance, and the certified lower optimum gain. Boundaries for these descriptive bins are declared in the analysis script; they are not confidence-based subpopulation discoveries.

In the capacity-activation stratum every method activates root repair and attains the same decision value to numerical precision. Accordingly these rows show the power and cost of the constrained response, not a learned model's extrapolation. The older nonlinear shifted actors remain unfavorable and are reproduced in the immutable predecessor electronic companion, including their full diagnostic tables and failed matched-accuracy outcomes. That evidence is not erased by the new multistage result: the successful current construction learns resource prices for a response that retains all modeled constraints and switching, rather than directly transferring an unverified actor across nonlinear models.

CS.4.5 Verification scopes

The published research policies undergo 16,384 independent rational replays. The new multistage geometry suite contributes 128 further policy/certificate replays on 64 instances. Its numerical matrix residual bound is checked against certified reference brackets, not called an exact-arithmetic proof of the matrix inverse. The separate 64-case quartic suite verifies both the exact decomposition and the inexact-response inequality using rational arithmetic throughout. The written theorems, rather than these finitely many instances, establish the general mathematical claims. All follow-up timing pipelines must satisfy their requested full-objective certificate. Build checks separately verify manuscript inputs, cross-references, page layout, archival preservation, and the exact published revision SHA.

CS.5 Nonsmooth service decisions and frozen learning pipelines

The experiments separate structural computation, candidate quality, and complete deployment. They do not use the generality of a theorem to imply empirical coverage of every admissible model. The earlier noncentered, boundary-inclusive and nonlinear studies remain intact in the companion; this section exercises nonzero absolute switching and the new verified-cell construction.

CS.5.1 A declared service-control distribution and operational units

Consider binary scenario trees with discount factor 0.9 and equal branch probabilities. Thus $w_n = (0.9/2)^{\text{depth}(n)}$. Tiers lie in $[0, 1]$, the installed tier is $1/2$, and external net-payment coefficients are $c_n = w_n[0.8 + 0.1(n \bmod 5)]$, with nodes numbered from zero. The accepted polytope includes every non-root continuation cap and the root payment equality. Physical service and stock are held at their identified

benchmark levels; the experiment concerns the remaining tier decision, not an untested inventory-learning claim.

The incremental profit, in profit units, is

$$G_{\alpha,\beta,\lambda}(x) = \sum_n (g_{0n} + \alpha g_{1n} + \beta g_{2n})(x_n - \frac{1}{2}) - \sum_n w_n (x_n - \frac{1}{2})^2 - \frac{1}{10} \sum_{n \neq o} w_n (x_n - x_{pa(n)})^2 - \lambda \sum_n w_n |x_n - x_{pa(n)}|, \quad (\text{CS.1})$$

where the root's predecessor is $1/2$ in the last term. Three fixed vectors describe net marginal resource offsets after the tariff; each has zero sum. Their exact rational coordinates and generating seed are recorded. Therefore the *reoptimized* fixed-tier comparator is exactly $1/2$ for every context: a common tier change has zero linear gain and strictly positive quadratic cost. This balanced-shock class is different from, and does not replace, the earlier noncentered class with boundary comparators.

One profit unit is the coefficient of a squared full-tier deviation in the per-review maintenance cost. The smooth inter-review cost has coefficient 0.1, and λ prices an absolute unit of reconfiguration. Adding a common level cost can make all gross resource costs positive without changing decisions. Results are divided by the discounted review mass $S_0 = \sum_n w_n$. Multiplying all monetary primitives by a currency conversion multiplies every profit and certificate by that conversion; no such conversion is a field calibration. For example, a gain of 0.003 represents 0.30 per discounted review if one profit unit is defined as 100 currency units. No operational population is asserted to have that calibration.

CS.5.2 Friction changes decisions, not only optimality labels

We solve 81 explicit epigraph quadratic programs: three contexts, trees with 15, 31, and 63 nodes, and nine friction values spanning zero to 1.25 times the computed critical value. Table 18 shows one context; every policy, tension, continuation price, and exact gap remains in the data. Nonzero friction changes both binding continuation constraints and fused edges. Switching decreases along the path and accepted gain disappears at the critical value to the certified tolerance. The zero-friction row is a control; the learning distribution below uses strictly positive friction.

The critical-friction study also includes binary and star trees from 31 to 1,023 nodes. Table 19 compares the factored and materialized linear programs using the same solver. On a star, forming *DB* creates quadratically many entries, while the two-factor representation retains $9N - 3$ nonzeros. Binary trees do not have this fill-in, and materialization can be faster there. The independent first-order algorithm produces rigorous brackets, but the 12,000-iteration cap does not always attain its 10^{-4} width target. Those failures remain visible. The result is a sparsity and certification consequence of the continuation structure, not a blanket speed claim for a custom optimizer.

CS.5.3 Learning, verification, and actual refinement

The learning experiment uses 31-node problems. Context coordinates are independent uniforms on the rational grid $\{-1, -0.999, \dots, 1\}$; friction is independently uniform on $\{0.01, 0.011, \dots, 0.20\}$. We solve

1,024 training contexts, recover candidate bases, and retain 747 distinct verified cells. A 192-unit random-feature ReLU layer, with a fitted multiclass ridge readout, proposes cell indices. Its scalar experts and value gradients are given by Theorem EC.7.5. The classifier learns joint continuation and switching patterns; it is not the earlier derivative-weighted regressor, and no superiority of derivative supervision is inferred.

We freeze three static-gated policies: check the first neural proposal, check the first three neural proposals, or check the cell of the nearest training anchor. Each returns the first region-valid proposal after exact flow repair and certificate calculation, comparing its reward with static. If no candidate survives, it returns static. Validation does not enlarge the library, update features, or tune a stopping rule. The complete model hash is recorded before and after both validation cohorts.

A separate 48-context matched-accuracy test adds actual classical refinement whenever the selected policy has normalized gap above 10^{-7} . It includes a cold epigraph quadratic program and sequential checking of the same frozen cell library. Table 20 includes generation, membership, exact audit, and all refinement time. The first-proposal face accuracy and added iteration counts distinguish learned candidate quality from a classical rescue. The data also contain raw face masks, switching signs, distances, and the acceptance and switching leakage of 96 neural/nearest proposals. Finite differences check the scalar experts' context-gradient pullbacks; these numerical checks supplement, not replace, the envelope proof.

Offline construction and fitting cost 18.22 seconds in the recorded environment. Under the observed mean timing difference, neural lookup recovers this cost after 4348 repeated queries relative to cold optimization; this is an accounting calculation, not a forecast or an advantage over sequential cell checking. The small, fixed-dimensional context distribution makes such reuse possible. The cell library can grow rapidly with problem complexity. Timings are single-environment descriptive measurements, and sequential checking is the stronger latency comparator in this run. The positive learning result is verified reuse with explicit rejection and acceptance guarantees, not a claim that the neural router dominates classical lookup.

CS.5.4 Executed independent validation, including a useful gain certificate

A first cohort of 4,096 independent contexts uses the uniform range from the outside gradient residual. Its broad expectation bounds are retained in the data. We then fix a model-derived range using the four deterministic forcing corners at the minimum friction, without altering any policy. The confirmation plan is saved before a second, independent 4,096-context cohort is drawn. This prospective second cohort is conditional on the already frozen model and the declared range calculation; it is not a refit on the first cohort.

Table 21 executes Proposition EC.7.6 with $M = 3$, $n = 4096$, and $\delta = 0.05$. The selected three-cell neural policy has empirical gain 0.003006 and a simultaneous expected-gain lower bound of 0.001754 per discounted review relative to static. The normalized expected-regret upper bound is 0.036684. The gain guarantee is positive at the economic improvement scale, whereas the regret ceiling is too wide to establish near-optimality. Classical refinement, separately timed above, is needed for the high-accuracy deployment guarantee. No significant advantage over the nearest-anchor policy or over a fully optimized policy follows from this table.

CS.5.5 Retained evidence and independent audit

The noncentered nonlinear study remains scientifically material: 48 instances, 432 deployments, boundary comparators, severe raw actor failures, and slower learned pipelines than cold SLSQP at matched accuracy. Its full text and tables are in Section EC.8; no unfavorable row is removed or pooled with the new distribution. The first four frozen predecessor seeds were an economical transfer diagnostic, not a representative sample of training variability. The inconclusive derivative-weighting comparison remains inconclusive. The earlier 992-coordinate failures and their actually executed fallbacks also remain available.

The new standard-library-only verifier independently reconstructs 10,312 rational policy certificates, 320,104 continuation checks, all corner ranges, and 12 rational critical-friction brackets. It imports neither the optimizer nor a numerical linear-algebra library. Optimizer flags, approximate face masks, and hardware timings are not treated as certificates. All data are synthetic; the mathematical scope, fixed-context validation scope, and previous transfer evidence are kept distinct.

CS.6 Direct scalar-critic deployment and independent confirmation

This study exercises Theorem EC.8.2 without an offline face library or a query-time optimal policy. It complements, rather than pools with, the preceding multistage experiment. A star has two review dates and many possible continuations; its large leaf count is not evidence about increasing tree depth.

CS.6.1 Primitives, frozen design, and unrefined decisions

There are 127 equally likely continuations and discount 0.9, giving root weight one, leaf weight $w = 0.9/127$, and total discounted review mass 1.9. Tiers lie in $[0, 1]$ with outside tier $1/2$, $\chi_0 = 2$, and $\chi_i = 2w$. All absolute switching weights equal their review weights. The root tariff is one; leaf tariff coefficients are $wk_i/8$, with fixed integers k_i drawn from $8, \dots, 12$. Balanced rational resource offsets make $1/2$ the reoptimized common-tier comparator in every context. The companion specifies the full generating formula and seed, not only the reduced objective. Profits and certificates below are divided by 1.9; gradient errors refer to the unnormalized value and the stated unit of h .

Training and test contexts have independent coordinates: h and α are uniform on the thousandth grid in $[-1, 1]$, and λ is uniform on the thousandth grid in $[0.01, 0.20]$. Thus every deployment has strictly positive absolute friction. Eight independent training seeds each generate 1,024 contexts and a 192-unit random-feature tanh scalar critic with a fitted ridge readout. Within each seed, value-only and value-gradient fits share the examples and hidden features. Their squared-error objectives differ only by the prespecified weight 0.25 on the squared h -derivative error; the ridge coefficient is 10^{-9} . The frozen design, coefficients, labels, and raw per-seed observations are retained. This is a prospective computational plan within the revision, not an externally registered clinical or field protocol.

An independent shared cohort of 128 contexts measures unrefined proposals. Table 22 reports all 2,048 policies. Mean optimum gain is 0.122047; value-only and value-gradient proposal regrets are 5.965×10^{-6} and 2.210×10^{-6} . Every proposal passes the exact gain gate; root-capacity projection occurs on 1.56% of

proposals. The derivative-weighted mean is better, but its paired regret difference has a 95% Student interval $[-9.568 \times 10^{-6}, 2.059 \times 10^{-6}]$, which includes zero. The unit of replication is the training seed, not the 128 reused contexts. This interval is conditional on that test cohort and does not establish a population-wide derivative-supervision advantage.

The observable residual and partial-gradient bounds are small on the profit scale, unlike a condition requiring an unknown optimal face. Exact certification is stricter: only 17.48% and 19.04% of the respective proposals immediately meet the normalized 10^{-7} tolerance. Table 23 in the companion charges prediction, capacity repair, rational auditing, and every executed refinement, with cold scalar Newton, Brent, and sorted breakpoints as strong structure-aware comparators. In this environment cold Newton’s median complete time is 5.353 ms and the value-gradient pipeline’s is 10.600 ms. The learned pipeline is slower, despite fewer refinement iterations. Its positive result is accurate, accepted *unrefined* deployment and a usable value-gradient theorem, not a latency claim obtained by omitting audits or choosing only a dense baseline.

CS.6.2 An executed expected-gain guarantee without classical rescue

After freezing the two critics from the first prespecified training seed, we draw a separate 2,048-context cohort with a different recorded seed. Neither model is refit or selected using this cohort before evaluation. Both policies use the feasible price response and exact static gate, with no optimization of the full query objective. Because $G_\lambda(y)$ is increasing in h and decreasing in λ for every $y \in Y$, and its maximum is convex in α , the optimum over the entire context box is bounded by two deterministic corners: $h = 1$, $\lambda = 0.01$, $\alpha \in \{-1, 1\}$. Exact conjugate upper bounds at those corners give a uniform normalized gain range $[0, 0.382478]$ for every gated policy.

A one-sided Hoeffding bound and a union bound over $M = 2$ frozen policies therefore give simultaneous expected-gain lower bounds with $n = 2048$ and $\delta = 0.05$. The selected value-gradient policy has empirical gain 0.119594 and lower bound 0.108115 per discounted review. Its average exact suboptimality certificate is 1.374×10^{-6} ; this last number is an empirical average, not a confidence bound on expected regret. Both policies have positive gain lower bounds, and their small gain difference is not established as significant. This confirmation concerns the declared fixed synthetic distribution, not arbitrary shift or new primitive families. It does demonstrate positive accepted neural deployment without the classical rescue needed by the earlier poor actors.

CS.6.3 Structural cross-checks and preservation of adverse evidence

The direct threshold agrees with 160 general tension linear programs. Across 480 solver comparisons, safeguarded Newton, Brent, bisection, and the breakpoint scan agree; 128 small-instance SLSQP comparisons have maximum value discrepancy 1.485×10^{-13} . A further 800 arbitrary-price checks include 106 root-capacity-binding cases and verify both global bounds and their underlying remainder identity. These are implementation tests, not substitutes for the proofs. Timing the explicit threshold and scalar solvers through 32,767 leaves confirms the intended sparse implementation; the raw measurements and their two-date limitation appear in Table 24.

A separate standard-library verifier reconstructs 6,626 rational policy certificates and 2,577,514 exact algebraic checks, including every continuation cap, root equality, absolute switching identity, profit, and conjugate upper bound. The full nonlinear and boundary-inclusive predecessor experiment, its badly shifted actors, unresolved derivative comparison, and slower complete learned pipelines remain unchanged. The new result identifies a tractable institution and learning statistic under which the original integrated research objective succeeds; it does not erase failure in the broader distributions.

CS.7 Implementation details for verified continuation cells

All essential new proofs are in the main paper. This section specifies the algorithms sufficiently to reproduce their numerical implementation. The ordinary Python sources contain no service calls and use pinned NumPy and SciPy versions. Training, validation, confirmation, structural tests, and matched deployment use separate recorded seeds. The confirmation stage does not change the frozen trained model.

The analytical sparse bound remains in Electronic Companion Section EC.9.

CS.7.1 Cell recovery and rejection

Training starts with an explicit epigraph quadratic program. A candidate active set is only a proposal. Pivoted factorization removes redundant equality rows, and a dual feasibility linear program recovers nonnegative inequality prices and capacity-feasible switching tensions. Positive inequality prices and unsaturated zero-edge tensions supply a candidate independent basis. Saturated zero-edge tensions may instead be represented by a signed cell. Every resulting affine map is checked against all primal and dual region inequalities. Failure to find a verified basis is recorded rather than assigned a fabricated label.

The input coordinates are scaled by $(1, 1, 10)$. The random hidden weights are standard normal; biases are uniform on $[-1, 1]$. Features concatenate a constant, the three scaled inputs, and 192 ReLU outputs. The multiclass readout is fitted by ridge least squares with coefficient 0.01 against one-hot cell labels. All random features and coefficients are serialized. The neural policy checks its ordered first one or three cells. The nearest-anchor comparator uses Euclidean distance in the same scaled context coordinates and then checks that anchor’s cell. Sequential lookup tests the same cells in their fixed training order. It receives no different library or optimal query-time face.

Numerical membership permits a 2×10^{-9} inequality tolerance solely as a proposal filter. The deployed flow is rounded to denominator 10^9 , negative flow coordinates are set to zero, and a common radial scaling enforces both tier bounds. Exact subtree feasibility survives this scaling. Nonnegative prices and capacity-feasible tensions are then rounded/clipped, and Theorem EC.7.4 is evaluated in rational arithmetic. Thus the numerical region tolerance is not the final feasibility or optimality tolerance. Wrong switching signs contribute leakage, including when rounding opens a nominally fused edge.

CS.7.2 What each timing and face statistic measures

The matched test measures complete pipelines, including exact audit and any classical rescue. Rescue uses a cold feasible start; a rejected region is not credited with unexecuted warm-start savings. The data separate its refinement iterations from initial cell verification. Distance is measured from the first affine proposal to the separately certified reference policy. Binding-constraint masks and switching signs use a 10^{-6} diagnostic tolerance and are not proofs that a mathematically active face has been identified. The independent verifier instead checks their associated repaired policies and all prices. The first sequentially tested cell is fixed, so its first-face accuracy has a different interpretation from the learned first proposal.

Offline amortization divides library-construction plus fitting time by the observed mean cold-minus-neural online time, when positive. It excludes no observed online audit cost, but it does not include hypothetical future library maintenance. It also does not turn one timing sample into a confidence statement. The complete matched records allow recalculation under a different deployment volume or hardware cost.

CS.7.3 Independent sampling and range reconstruction

Training uses seed 13031 and random features seed 13032. The first validation uses 13033 and matched deployment 13034. After the deterministic corner range is saved, confirmation uses 13035. Within each cohort the two forcing coordinates and friction are independently generated from the specified finite uniform grids. Samples are not resampled when routing fails: the prescribed static policy is retained, and its certificate contributes to the mean. The first broad-range cohort remains in full. The second range is derived from the model’s four forcing vertices, not from truncating unfavorable sample values. For each vertex the standard-library verifier checks both its policy and its upper bound before accepting the range.

The final confidence statement is conditional on the frozen model, distribution, and monetary normalization. Its positive gain floor compares with the static outside arrangement. The simultaneous regret ceilings are intentionally reported alongside it: they remain much wider than high-accuracy deployment tolerances. This distinction prevents the validation theorem from being described as a blanket certification of neural training, distribution shift, or superiority over classical control.

CS.8 Reconstruction and execution details for the scalar-critic study

CS.8.1 Primitive service instance and comparator

Let $m = 127$, $w = 9/(10m)$, $k_i \in \{8, \dots, 12\}$, $u_i \in \{-28, \dots, 4\}$, and $v_i \in \{-20, \dots, 20\}$. NumPy’s default generator with seed 14001 draws these three integer arrays in that order. They are also stored explicitly, so replay does not depend on a generator implementation. Put $\rho_i = 9k_i/(80m)$, $A = -w \sum_i u_i/20$, $B = -w \sum_i v_i/20$, and $r = \sum_i \rho_i$. Smooth reward gradients at the common outside tier are

$$g_0 = A + \alpha B + \frac{hr}{1+r}, \quad g_i = \frac{w(u_i + \alpha v_i)}{20} - \frac{h\rho_i}{1+r}.$$

Their sum is exactly zero, and $\rho_i g_0 - g_i = \rho_i A - w u_i/20 + \alpha(\rho_i B - w v_i/20) + h \rho_i$. A common tier displacement therefore has zero first-order reward and strictly positive quadratic cost; installation friction reinforces that inequality. Thus $1/2$ is the reoptimized static comparator, not an arbitrary fixed baseline. The smooth node reward is $g_i(x_i - 1/2) - \chi_i(x_i - 1/2)^2/2$ relative to that baseline. Given fixed tariff c_i , take its resource increment to be $c_i(x_i - 1/2)$ minus this reward. Adding a sufficiently large constant makes all gross resource costs nonnegative on the compact tier interval without changing any decision. This is the identified-stock, quadratic-resource specialization of the unified model, not an empirical fit of demand or a claim that every inventory cost reduces to a quadratic.

CS.8.2 Neural loss, actual computation, and reproducibility

Write $z = (h, \alpha, (\lambda - 0.105)/0.095)$ and $\widehat{J}_\omega(z) = a + b^\top z + \sum_{j=1}^{192} c_j \tanh(W_j^\top z + \zeta_j)$. The hidden coefficients are fixed normal draws with standard deviation 1.5, and biases are uniform on $[-1, 1]$. Only a, b, c are fitted. The derivative of this same scalar function, not an independent vector actor, produces the coupling price. Value and value-gradient fits minimize the sample mean of $(\widehat{J} - J)^2 + \gamma(\partial_h \widehat{J} - v^*)^2$ plus 10^{-9} times the squared readout norm, with $\gamma = 0$ and 0.25 , respectively. The analytical derivative is used in training and deployment. Targets use the structure-aware scalar solve; small-instance independent optimization and exact gap replay check the solver. In finite precision target optimality is established to the recorded numerical tolerances, not as a symbolic expression for every training label.

The paired seed interval in the paper is a Student interval on eight seed-level mean differences. It is a conventional small-sample approximation, not a distribution-free guarantee. The independent gain bound is different: its bounded-variable argument is conditional on the frozen models and uses no Gaussian approximation. Only the primary paired regret endpoint was designated before fitting; secondary comparisons are descriptive and do not receive unadjusted claims of significance.

The total label-generation time, charged once per shared training sample rather than twice for the paired methods, is 0.756 seconds; fitting the eight value-only and eight derivative-weighted readouts takes 0.091 and 0.076 seconds, respectively. These offline costs are not included in per-query timings and are not claimed to amortize into a speed advantage. Raw deployment timings cover 32 fixed test contexts, three repetitions, and a rotated method order. The exact audit is performed for every proposed policy and again after any refinement. Components and total times are retained for every repetition, including immediate policies. No failed proposal or timing row is dropped.

The scaling experiment varies branching at a fixed two-date horizon. The threshold requires one vector scan. Breakpoint sorting gives a worst-case arithmetic bound; the implementation additionally benchmarks safeguarded Newton and Brent because they can be faster in practice. The table times the structural calculation itself, not a complete audited deployment; its heading and the separate complete-timing table make that distinction explicit. Five raw repetitions, iteration counts, and primitive seeds are retained at each size.

CS.8.3 Independent certificates and exact fallback semantics

The proposer rounds each leaf transfer down to a multiple of 10^{-9} , clips to its interval, and rescales by an exact common rational factor if the root cap is exceeded. This preserves all leaf continuation inequalities and the root equality by construction. It rounds a proposed total dual price to a rational number and evaluates every conjugate directly. Such a price need not satisfy approximate stationarity: weak duality alone validates the upper bound. The independent verifier imports only the Python standard library, reconstructs all primitive coefficients from the stored integers, and checks exact rational equalities, inequalities, absolute switching, profit, and upper bounds. Numerical gradient inequalities and wall-clock times are explicitly outside that rational proof checker.

The validation policy performs no full objective solve after prediction. The capacity-only response can require scalar water filling, which is part of feasibility implementation and is not hidden as a free oracle. The matched-accuracy policy is a different, explicitly labeled pipeline: a certificate gap above 10^{-7} invokes the full scalar solver and a second audit. Its last policy is feasible and its exact gap meets that tolerance. The raw-policy table and independent expected-gain guarantee never substitute this refined policy for the learned proposal.

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Table 1: Separate offline information and predictor storage costs. Label time is the full cost for a standalone deployment of either method. Reachable array bytes are not a peak-memory measure.

Nodes	Method	Labels (s)	Fit (s)	Serialized bytes	Array bytes
63	Tanh gradient	0.852	0.03149	34581	13888
63	RBF direct	0.852	0.00026	11536	14936
127	Tanh gradient	5.852	0.00307	57271	23152
127	RBF direct	5.852	0.00025	21982	29288
255	Tanh gradient	18.536	0.00147	34485	13888
255	RBF direct	18.536	0.00020	11514	14936

Table 2: Matrix-input storage and whole-process peak memory. Sparse input bytes exclude solver factorization; peak RSS is a process high-water mark across methods, not an attributed method cost.

Nodes	Response input bytes	Lifted input bytes	Process peak KiB
63	26696	34964	107096
127	66772	99472	122884
255	119660	152660	168800

Table 3: Mean timing components in milliseconds at target 10^{-3} . Fallback columns are unconditional means; conditional means and iterations are in the machine-readable record.

Nodes	Method	Predict	Initial	Audit/polish	FB setup	FB solve	FB audit
63	Lifted continuation	0.00	3.48	2.59	0.00	0.00	0.00
63	Tanh gradient	0.07	3.31	2.57	1.29	10.21	1.43
63	RBF direct	0.04	3.62	2.57	0.36	2.30	0.40
127	Lifted continuation	0.00	18.91	6.45	0.07	1.96	0.12
127	Tanh gradient	0.08	15.62	6.51	2.17	83.34	3.80
127	RBF direct	0.07	17.22	6.53	2.16	83.72	3.82
255	Lifted continuation	0.00	43.57	13.81	0.00	0.00	0.00
255	Tanh gradient	0.09	62.75	13.74	2.96	189.78	9.04
255	RBF direct	0.07	60.57	13.77	2.44	163.36	7.38

Table 4: Mean timing components in milliseconds at target 10^{-5} . Fallback columns are unconditional means; conditional means and iterations are in the machine-readable record.

Nodes	Method	Predict	Initial	Audit/polish	FB setup	FB solve	FB audit
63	Lifted continuation	0.00	3.45	2.57	0.52	3.18	0.58
63	Tanh gradient	0.06	3.28	2.56	1.28	10.19	1.43
63	RBF direct	0.04	3.32	2.57	1.32	10.19	1.43
127	Lifted continuation	0.00	19.04	6.53	1.40	66.80	2.42
127	Tanh gradient	0.08	16.05	6.51	2.16	83.83	3.84
127	RBF direct	0.07	17.40	6.52	2.17	83.57	3.83
255	Lifted continuation	0.00	44.19	13.76	2.40	172.09	7.41
255	Tanh gradient	0.09	61.25	13.80	2.94	190.74	8.95
255	RBF direct	0.07	58.47	13.75	2.95	190.68	8.94

Table 5: Fresh-host robust pointwise certification costs in milliseconds, including ensemble prediction. Sixteen contexts per radius. The entire comparator batch is charged once per independently certified method; its actual reuse in the experiment is disclosed.

Radius	Method	Prediction	Own total	Comparators	Complete total	Minimum gain
0.00	Protected tanh	0.31	25.08	122.00	147.08	0.741954
0.00	Protected RBF	0.22	20.46	122.00	142.46	0.742176
0.00	Robust maximin	0.00	28.14	122.00	150.14	0.742209
0.25	Protected tanh	0.32	23.62	134.52	158.14	0.276082
0.25	Protected RBF	0.23	23.45	134.52	157.97	0.296677
0.25	Robust maximin	0.00	43.63	134.52	178.15	0.413329

Table 6: Retrospective full R22 scaling variability, part 1. Specification is nodes/services/resources/capacities/curvature; C and S denote coarse and strict targets. Times and paired savings are milliseconds; each entry has only eight historical contexts.

Specification	Target	Method	Median	Quartiles	Savings	Resampling interval
31/2/2/4/1	C	Dense cold	6.61	[4.36, 7.05]	-1.83	[-2.60, -1.12]
31/2/2/4/1	C	Dense previous	5.63	[4.33, 6.21]	-1.20	[-1.44, -0.99]
31/2/2/4/1	C	Lifted continuation	4.33	[3.46, 4.84]	0.00	[0.00, 0.00]
31/2/2/4/1	C	Tanh gradient	4.62	[3.41, 5.30]	-1.28	[-3.68, 0.15]
31/2/2/4/1	C	RBF direct	4.59	[3.39, 4.81]	-0.04	[-0.41, 0.28]
31/2/2/4/1	S	Dense cold	6.40	[4.45, 6.92]	-1.89	[-2.89, -1.10]
31/2/2/4/1	S	Dense previous	5.73	[4.34, 6.49]	-1.23	[-1.45, -1.01]
31/2/2/4/1	S	Lifted continuation	4.43	[3.45, 4.93]	0.00	[0.00, 0.00]
31/2/2/4/1	S	Tanh gradient	10.40	[4.62, 14.17]	-4.22	[-7.02, -1.42]
31/2/2/4/1	S	RBF direct	4.71	[4.26, 6.20]	-0.95	[-2.74, 0.15]
63/2/6/4/1	C	Dense cold	14.79	[13.34, 19.16]	-7.53	[-10.33, -5.31]
63/2/6/4/1	C	Dense previous	15.15	[11.26, 17.12]	-6.58	[-8.17, -4.90]
63/2/6/4/1	C	Lifted continuation	7.73	[7.40, 8.85]	0.00	[0.00, 0.00]
63/2/6/4/1	C	Tanh gradient	27.98	[20.88, 34.07]	-19.14	[-25.97, -12.13]
63/2/6/4/1	C	RBF direct	11.80	[8.01, 25.25]	-8.60	[-17.55, -0.80]
63/2/6/4/1	S	Dense cold	15.91	[11.99, 38.57]	-10.54	[-20.48, -4.13]
63/2/6/4/1	S	Dense previous	17.20	[13.74, 27.72]	-6.70	[-8.35, -4.98]
63/2/6/4/1	S	Lifted continuation	9.36	[7.41, 17.84]	0.00	[0.00, 0.00]
63/2/6/4/1	S	Tanh gradient	26.95	[20.94, 32.27]	-13.89	[-19.79, -8.56]
63/2/6/4/1	S	RBF direct	26.85	[21.33, 33.47]	-14.33	[-21.39, -7.92]
127/2/6/4/1	C	Dense cold	89.17	[84.02, 150.56]	-82.57	[-132.75, -40.02]
127/2/6/4/1	C	Dense previous	93.27	[67.94, 143.38]	-69.86	[-107.42, -38.52]
127/2/6/4/1	C	Lifted continuation	30.17	[27.48, 32.66]	0.00	[0.00, 0.00]
127/2/6/4/1	C	Tanh gradient	158.46	[97.22, 202.14]	-162.09	[-294.10, -65.69]
127/2/6/4/1	C	RBF direct	98.77	[36.35, 175.85]	-76.60	[-136.79, -21.71]

Table 7: Retrospective full R22 scaling variability, part 2. Specification is nodes/services/resources/capacities/curvature; C and S denote coarse and strict targets. Times and paired savings are milliseconds; each entry has only eight historical contexts.

Specification	Target	Method	Median	Quartiles	Savings	Resampling interval
127/2/6/4/1	S	Dense cold	196.89	[145.35, 263.82]	-78.28	[-124.76, -48.79]
127/2/6/4/1	S	Dense previous	185.38	[155.70, 225.31]	-72.46	[-112.51, -40.07]
127/2/6/4/1	S	Lifted continuation	97.31	[27.56, 188.20]	0.00	[0.00, 0.00]
127/2/6/4/1	S	Tanh gradient	160.48	[97.42, 203.71]	-41.02	[-87.41, -4.06]
127/2/6/4/1	S	RBF direct	165.49	[102.48, 209.02]	-39.96	[-92.45, 1.84]
255/2/6/4/1	C	Dense cold	761.03	[516.41, 847.82]	-706.68	[-1061.96, -436.96]
255/2/6/4/1	C	Dense previous	322.15	[271.57, 490.48]	-425.23	[-737.41, -185.12]
255/2/6/4/1	C	Lifted continuation	77.54	[65.06, 103.02]	0.00	[0.00, 0.00]
255/2/6/4/1	C	Tanh gradient	340.97	[219.96, 396.40]	-254.82	[-346.71, -172.64]
255/2/6/4/1	C	RBF direct	349.44	[222.96, 392.78]	-271.56	[-385.45, -181.40]
255/2/6/4/1	S	Dense cold	947.41	[788.74, 1082.14]	-682.65	[-970.64, -443.55]
255/2/6/4/1	S	Dense previous	591.61	[433.13, 711.40]	-426.00	[-739.97, -214.63]
255/2/6/4/1	S	Lifted continuation	339.84	[164.58, 358.19]	0.00	[0.00, 0.00]
255/2/6/4/1	S	Tanh gradient	334.89	[221.29, 368.00]	-35.32	[-75.79, -0.73]
255/2/6/4/1	S	RBF direct	351.59	[224.05, 395.55]	-60.97	[-101.03, -23.52]
511/2/6/4/1	C	Dense cold	5672.76	[2275.67, 7021.47]	-4282.59	[-5855.53, -2710.38]
511/2/6/4/1	C	Dense previous	3709.47	[1375.41, 4139.87]	-2738.53	[-3982.86, -1653.14]
511/2/6/4/1	C	Lifted continuation	341.61	[150.87, 809.35]	0.00	[0.00, 0.00]
511/2/6/4/1	C	Tanh gradient	1848.06	[457.91, 2497.88]	-873.48	[-1624.04, -234.16]
511/2/6/4/1	C	RBF direct	1913.48	[489.98, 2544.60]	-902.41	[-1673.68, -265.04]
511/2/6/4/1	S	Dense cold	6980.97	[2749.97, 7923.67]	-3970.05	[-5084.99, -2766.39]
511/2/6/4/1	S	Dense previous	4268.82	[1593.52, 6203.13]	-2737.23	[-3949.99, -1702.51]
511/2/6/4/1	S	Lifted continuation	1824.31	[375.60, 2460.51]	0.00	[0.00, 0.00]
511/2/6/4/1	S	Tanh gradient	1804.50	[514.21, 2525.85]	-47.96	[-177.82, 84.30]
511/2/6/4/1	S	RBF direct	1921.35	[494.90, 2594.30]	-102.14	[-199.59, 1.63]

Table 8: Retrospective full R22 scaling variability, part 3. Specification is nodes/services/resources/capacities/curvature; C and S denote coarse and strict targets. Times and paired savings are milliseconds; each entry has only eight historical contexts.

Specification	Target	Method	Median	Quartiles	Savings	Resampling interval
1023/2/6/4/1	C	Dense cold	20995.42	[14990.50, 28234.16]	-20637.35	[-28400.62, -14990.43]
1023/2/6/4/1	C	Dense previous	14982.59	[11959.04, 20444.64]	-14546.06	[-20171.53, -10021.31]
1023/2/6/4/1	C	Lifted continuation	2482.09	[435.29, 5463.83]	0.00	[0.00, 0.00]
1023/2/6/4/1	C	Tanh gradient	3520.81	[1823.62, 5769.08]	-852.62	[-1272.19, -449.52]
1023/2/6/4/1	C	RBF direct	3669.73	[2077.82, 5643.88]	-955.14	[-1458.63, -460.73]
1023/2/6/4/1	S	Dense cold	20381.65	[15027.14, 27434.26]	-19716.70	[-27851.23, -13738.50]
1023/2/6/4/1	S	Dense previous	14999.09	[12318.13, 20403.96]	-13983.38	[-19565.17, -9056.35]
1023/2/6/4/1	S	Lifted continuation	3433.29	[1701.48, 5416.22]	0.00	[0.00, 0.00]
1023/2/6/4/1	S	Tanh gradient	3610.86	[1918.85, 5749.83]	-260.54	[-479.93, -78.38]
1023/2/6/4/1	S	RBF direct	3640.62	[1936.21, 5630.26]	-228.90	[-338.26, -120.01]
127/1/2/4/1	C	Dense cold	15.53	[13.66, 27.77]	-10.71	[-18.27, -4.27]
127/1/2/4/1	C	Dense previous	16.93	[12.01, 30.55]	-11.11	[-18.66, -4.30]
127/1/2/4/1	C	Lifted continuation	10.24	[7.32, 16.57]	0.00	[0.00, 0.00]
127/1/2/4/1	C	Tanh gradient	11.39	[8.85, 22.34]	-3.81	[-9.22, 1.19]
127/1/2/4/1	C	RBF direct	10.23	[8.78, 14.88]	-1.83	[-5.43, 1.74]
127/1/2/4/1	S	Dense cold	24.46	[20.19, 43.41]	-8.69	[-17.70, -0.96]
127/1/2/4/1	S	Dense previous	31.42	[15.58, 45.14]	-9.39	[-17.91, -1.66]
127/1/2/4/1	S	Lifted continuation	22.09	[15.78, 36.49]	0.00	[0.00, 0.00]
127/1/2/4/1	S	Tanh gradient	30.31	[19.87, 35.59]	-6.50	[-13.58, -0.07]
127/1/2/4/1	S	RBF direct	22.16	[19.00, 35.61]	-3.07	[-9.25, 3.46]
127/4/12/4/1	C	Dense cold	529.88	[426.90, 805.87]	-515.59	[-744.89, -324.03]
127/4/12/4/1	C	Dense previous	332.32	[276.44, 525.16]	-316.43	[-434.73, -209.80]
127/4/12/4/1	C	Lifted continuation	74.90	[58.16, 139.09]	0.00	[0.00, 0.00]
127/4/12/4/1	C	Tanh gradient	476.82	[364.49, 638.85]	-445.31	[-619.31, -297.79]
127/4/12/4/1	C	RBF direct	479.58	[369.44, 642.80]	-458.19	[-631.18, -312.21]

Table 9: Retrospective full R22 scaling variability, part 4. Specification is nodes/services/resources/capacities/curvature; C and S denote coarse and strict targets. Times and paired savings are milliseconds; each entry has only eight historical contexts.

Specification	Target	Method	Median	Quartiles	Savings	Resampling interval
127/4/12/4/1	S	Dense cold	892.98	[750.11, 1224.85]	-451.97	[-585.50, -311.72]
127/4/12/4/1	S	Dense previous	677.11	[581.51, 1154.37]	-349.94	[-463.76, -247.60]
127/4/12/4/1	S	Lifted continuation	452.74	[389.77, 653.53]	0.00	[0.00, 0.00]
127/4/12/4/1	S	Tanh gradient	499.87	[357.31, 632.83]	-14.49	[-96.17, 56.41]
127/4/12/4/1	S	RBF direct	479.15	[379.07, 645.77]	-31.82	[-128.62, 40.86]
127/2/12/8/1	C	Dense cold	112.63	[93.65, 200.25]	-66.34	[-113.41, -27.52]
127/2/12/8/1	C	Dense previous	129.75	[87.27, 179.76]	-54.61	[-81.08, -27.41]
127/2/12/8/1	C	Lifted continuation	51.60	[41.49, 82.58]	0.00	[0.00, 0.00]
127/2/12/8/1	C	Tanh gradient	134.98	[112.24, 165.08]	-63.97	[-121.80, -10.16]
127/2/12/8/1	C	RBF direct	134.11	[115.60, 178.85]	-68.40	[-122.32, -18.69]
127/2/12/8/1	S	Dense cold	210.87	[160.14, 265.91]	-46.56	[-71.50, -15.12]
127/2/12/8/1	S	Dense previous	230.19	[200.07, 285.75]	-69.57	[-90.24, -47.60]
127/2/12/8/1	S	Lifted continuation	140.67	[134.28, 192.11]	0.00	[0.00, 0.00]
127/2/12/8/1	S	Tanh gradient	128.85	[114.27, 164.72]	13.47	[-1.42, 28.97]
127/2/12/8/1	S	RBF direct	132.63	[107.03, 181.29]	8.87	[-5.59, 22.09]
127/2/6/4/16	C	Dense cold	90.11	[75.08, 102.92]	-63.00	[-77.44, -47.08]
127/2/6/4/16	C	Dense previous	61.75	[55.24, 73.79]	-44.25	[-54.16, -36.03]
127/2/6/4/16	C	Lifted continuation	22.71	[21.05, 23.43]	0.00	[0.00, 0.00]
127/2/6/4/16	C	Tanh gradient	88.87	[73.25, 101.08]	-69.70	[-92.24, -51.03]
127/2/6/4/16	C	RBF direct	67.09	[48.15, 101.02]	-54.18	[-82.66, -28.18]
127/2/6/4/16	S	Dense cold	145.26	[75.46, 160.36]	-62.58	[-77.38, -46.46]
127/2/6/4/16	S	Dense previous	114.41	[69.04, 126.59]	-44.58	[-53.70, -37.30]
127/2/6/4/16	S	Lifted continuation	67.29	[23.40, 93.13]	0.00	[0.00, 0.00]
127/2/6/4/16	S	Tanh gradient	86.95	[71.49, 99.93]	-20.78	[-37.28, -6.84]
127/2/6/4/16	S	RBF direct	84.46	[69.70, 100.48]	-21.12	[-37.59, -7.17]

Table 10: Coarse-target scaling on complete accepted polyhedra (mean milliseconds).

Nodes $\times$ services	$r/c/q$	Cold	Previous	Lifted	Tanh	RBF
31×2	2/4/1	6.09	5.47	4.26	5.54	4.31
63×2	6/4/1	16.40	15.45	8.87	28.01	17.48
127×2	6/4/1	119.22	106.52	36.65	198.75	113.26
255×2	6/4/1	794.76	513.31	88.07	342.89	359.63
511×2	6/4/1	5425.71	3881.65	1143.13	2016.61	2045.53
1023×2	6/4/1	23790.22	17698.93	3152.87	4005.49	4108.01
127×1	2/4/1	22.46	22.87	11.76	15.57	13.58
127×4	12/4/1	619.17	420.01	103.58	548.89	561.77
127×2	12/8/1	142.29	130.56	75.95	139.92	144.35
127×2	6/4/16	85.81	67.07	22.81	92.51	77.00

Each entry includes eight held-out complete attempts. The second column reports resource rank, additional capacities, and the alternating curvature multiplier. All methods receive the same independent target, price repair and actual fallback. Parenthesized superscripts, when present, count failed target checks; such attempts remain in the recorded cost and do not establish a certified speed advantage. A separate 64-label fit is used at every configuration.

Table 11: Actual geometry in the new independent validation populations.

Domain	Contexts	Out of support (%)	Capacities	Tier bounds	Kinks
IID	2048	0.00	0.000	0.063	61.749
SHIFT	512	93.55	0.000	0.178	65.424

The last three columns are mean counts in the audited full reference decision. Zero capacity activation is reported as zero, not relabeled as a capacity stress test. The deliberate inherited R19 suite remains the separate test of forced active geometries.

Table 12: Measured scaling offline work and formulation sizes.

Nodes $\times$ services	Labels (s)	Fits (s)	Dense setup	Lifted setups	Dense P	Lifted P
31×2	0.448	0.034	0.001	0.001	3,566	64
63×2	1.747	0.003	0.002	0.002	15,464	132
127×2	4.489	0.003	0.016	0.003	62,804	260
255×2	16.793	0.002	0.045	0.006	253,474	516
511×2	96.487	0.003	0.227	0.008	1,017,116	1,028
1023×2	232.009	0.002	1.506	0.016	4,077,840	2,052
127×1	1.772	0.002	0.002	0.001	14,761	129
127×4	20.889	0.003	0.074	0.006	253,614	520
127×2	7.969	0.003	0.016	0.003	63,432	266
127×2	4.653	0.003	0.015	0.003	62,804	260

Labels include 64 lifted full-objective solves and independent audits. Fits include both declared predictors. Lifted setup charges two tolerance-isolated workspaces; dense cold setup charges one. The last columns report stored quadratic-matrix nonzero counts, not an asymptotic complexity theorem. Primitive construction and all remaining setup timings are retained in the raw summary.

Table 13: Unrefined multistage decisions and conditional frozen-fleet gains

Method	Regret upper	Mean gap	Pass (%)	Fleet lower
Tanh: value	0.034374	0.193705	0.00	0.542816
Tanh: value + gradient	0.003685	0.015856	0.54	0.573506
Tanh: price	0.001581	0.007986	3.66	0.575609
Quadratic: value	0.003079	0.012446	4.59	0.574111
Quadratic: value + gradient	0.002794	0.011081	9.47	0.574396
Quadratic: price	0.000271	0.001238	65.48	0.576919
RBF: price	0.000158	0.000657	86.43	0.577032

All 2,048 independent deployment contexts per method are included. Pass means the immediate exact full-objective gap is at most 10^{-3} . No method passes at 10^{-5} before refinement. Fleet bounds apply to the explicitly randomized eight-model deployment, not each model or arbitrary retraining.

Table 14: Deliberate multistage geometry checks

Stratum	Cases	Capacity	Tier boundary	Kinks	Gradient error
heterogeneous	16	0	0	16	1.04e-06
binding capacity	16	16	0	16	5.10e-08
tier boundary	16	0	16	16	1.48e-11
switching kinks	16	0	0	16	7.47e-05

Capacity, tier boundary, and kinks count instances with at least one such event; diagnostics use 10^{-5} . Each of 128 implemented reference/proposal certificates is independently replayed. The gradient column is the maximum finite-difference error among the four checked contexts in that stratum.

Table 15: Accuracy-targeted complete-pipeline cost

Method	Mean ms 10^{-3}	Mean ms 10^{-5}	Fall back (%)	Fall back (%)
Classical: cold	14.029	26.180	0.00	42.19
Classical: previous	13.453	26.822	0.00	43.36
Tanh: value	43.712	43.532	100.00	100.00
Tanh: value + gradient	42.169	42.721	99.22	100.00
Tanh: price	41.519	41.594	98.44	100.00
Quadratic: value	40.264	41.201	95.31	100.00
Quadratic: value + gradient	39.654	42.080	91.41	100.00
Quadratic: price	19.790	39.858	44.14	100.00
RBF: price	9.563	37.808	12.89	100.00

The last two columns correspond to 10^{-3} and 10^{-5} , respectively. All methods start with numerical tolerance 10^{-5} , then refine and re-audit when the independent full-objective gap fails the requested target. Each mean uses 256 records. No prediction, response, repair, audit, or actual refinement cost is omitted.

Table 16: Timing dispersion and conservative offline break-even

Method	Target	Median ms	95th percentile ms	Break-even
Classical: cold	10^{-3}	13.50	25.40	—
Classical: cold	10^{-5}	16.73	84.91	—
Classical: previous	10^{-3}	12.54	25.53	—
Classical: previous	10^{-5}	17.04	83.12	—
Tanh: value	10^{-3}	34.32	109.65	—
Tanh: value	10^{-5}	33.58	109.61	—
Tanh: value + gradient	10^{-3}	31.22	100.77	—
Tanh: value + gradient	10^{-5}	32.29	101.52	—
Tanh: price	10^{-3}	29.29	102.34	—
Tanh: price	10^{-5}	29.23	104.87	—
Quadratic: value	10^{-3}	30.13	98.88	—
Quadratic: value	10^{-5}	30.51	98.70	—
Quadratic: value + gradient	10^{-3}	28.97	102.22	—
Quadratic: value + gradient	10^{-5}	30.95	102.02	—
Quadratic: price	10^{-3}	9.57	58.39	—
Quadratic: price	10^{-5}	29.39	99.10	—
RBF: price	10^{-3}	6.27	28.49	2546
RBF: price	10^{-5}	29.50	96.46	—

A dash means no positive observed saving (or a classical method), not unmeasured overhead. All same-host label-generation and seven-model fitting cost is charged to each surrogate. Raw rows, component costs, and approximate eight-cohort timing-difference intervals are supplied in the results file.

Table 17: Retained star stress strata: selected columns, complete records supplied

Stratum	Method	Regret upper	Root repair (%)
original	Tanh: value	2.85e-06	0.0
original	Tanh: value + gradient	2.15e-07	0.0
original	Tanh: price	1.74e-07	0.0
original	Quadratic: price	1.91e-04	0.0
original	RBF: price	4.41e-08	0.0
critical-friction	Tanh: value	1.07e-07	0.0
critical-friction	Tanh: value + gradient	1.09e-08	0.0
critical-friction	Tanh: price	6.96e-09	0.0
critical-friction	Quadratic: price	4.69e-06	0.0
critical-friction	RBF: price	5.54e-09	0.0
capacity-activation	Tanh: value	1.48e-10	100.0
capacity-activation	Tanh: value + gradient	1.48e-10	100.0
capacity-activation	Tanh: price	1.48e-10	100.0
capacity-activation	Quadratic: price	1.48e-10	100.0
capacity-activation	RBF: price	1.48e-10	100.0
primitive-perturbation	Tanh: value	1.36e-03	0.0
primitive-perturbation	Tanh: value + gradient	1.35e-03	0.0
primitive-perturbation	Tanh: price	1.35e-03	0.0
primitive-perturbation	Quadratic: price	1.43e-03	0.0
primitive-perturbation	RBF: price	1.35e-03	0.0

All eight methods, all four seeds, every stratum, and all geometry bins are retained in the machine-readable results; the table displays five methods for readability. Every method in the capacity-activation stratum has the same regret bracket to numerical precision. These are existing R16 observations, not newly generated R19 evidence.

Table 18: Accepted decisions along nonzero switching-friction paths

Nodes	λ/λ_c	λ	Gain	Switching	Binding caps	Fused edges
15	0.00	0.0000	0.008837	0.08549	8	0
15	0.50	0.2066	0.001294	0.01253	13	12
15	1.00	0.4131	0.000000	0.00000	14	15
15	1.25	0.5164	0.000000	0.00000	14	15
31	0.00	0.0000	0.014286	0.09843	14	0
31	0.50	0.2347	0.002270	0.02281	22	15
31	1.00	0.4694	0.000000	0.00000	30	31
31	1.25	0.5867	0.000000	0.00000	30	31
63	0.00	0.0000	0.015029	0.11327	26	2
63	0.50	0.2823	0.001509	0.01341	45	41
63	1.00	0.5646	0.000000	0.00000	62	63
63	1.25	0.7058	0.000000	0.00000	62	63

Context (0.2, -0.4); all 81 paths across three contexts are retained in the data. Gain and switching are per discounted review. Binding and fused masks use 10^{-6} for numerical diagnosis; exact rational policy gaps, rather than masks, determine validity. The zero-friction row is the control, not the learning experiment.

Table 19: Factored critical-friction computation and independent brackets

Tree	Nodes	Factored entries	Expanded entries	Factored ms	Expanded ms	Bracket width
Binary	31	276	242	2.26	2.56	0.00003
Binary	127	1,140	1,010	3.83	3.74	0.00009
Binary	1023	9,204	8,178	33.63	23.97	0.00674
Star	31	276	1,054	1.55	2.63	0.00008
Star	127	1,140	16,510	3.43	7.03	0.00008
Star	1023	9,204	1,049,598	51.70	462.99	0.00175

Both linear programs use HiGHS; expanded time includes materializing *DB*. The final column is the exact rational width from the separate accelerated projected-gradient algorithm, capped at 12,000 iterations and targeting 10^{-4} . Widths above target are retained, not recorded as successful solves. Sparse arithmetic is not a claim that this first-order algorithm beats the linear-programming solver.

Table 20: Complete deployment at a common certified gap

Method	Milliseconds	Immediate (%)	Refine (%)	First face (%)	Added iterations
Cold quadratic program	19.74	—	—	—	—
Neural, three cells	17.94	31.2	68.8	16.7	-7.90
Nearest anchor	19.82	20.8	79.2	22.9	-5.21
Sequential cell check	5.83	52.1	47.9	0.0	-13.27

Forty-eight fresh 31-node problems. Time includes proposal generation, membership checking, exact rational audit, and any executed refinement. Refused cells use a cold classical refinement, not an unreported warm start. All final normalized gaps are at most 10^{-7} . First-face masks are descriptive and use the tolerance stated in the data. Offline construction is reported separately; these timings are hardware-dependent observations, not population speed claims.

Table 21: Independent validation of frozen, static-gated pipelines

Pipeline	Cell accepted (%)	Mean gain	Expected gain floor	Expected regret ceiling
Neural, one cell	16.43	0.001734	0.000482	0.042906
Neural, three cells	28.86	0.003006	0.001754	0.036684
Nearest anchor	21.34	0.002291	0.001039	0.040445

A separate sample of $n = 4096$ contexts, three frozen pipelines, and $\delta = 0.05$. Both expectation bounds hold simultaneously under the declared deployment distribution. The model-derived range is $B = 0.051784$ and the deviation radius is 0.001252. The gain floor compares with static, not with an optimized nonstatic policy. The regret ceilings are valid but not small enough to imply near-optimality.

Table 22: Unrefined accepted scalar-critic policies: eight paired training seeds, 128 shared independent contexts, and positive switching friction.

Metric	Value only	Value gradient
Mean normalized regret	5.965×10^{-6}	2.210×10^{-6}
Mean exact normalized gap	1.534×10^{-5}	5.367×10^{-6}
Mean residual bound	9.131×10^{-6}	3.194×10^{-6}
Mean gradient bound	2.224×10^{-5}	7.046×10^{-6}
Partial-gradient squared error	1.171×10^{-5}	3.708×10^{-6}
Immediately certified at 10^{-7}	17.48%	19.04%
Negative gain, static gate	0.00%	0.00%

All proposal losses and bounds precede full-objective refinement. The paired primary regret interval includes zero; lower mean error alone does not establish a derivative-training advantage. The exact gap includes rational repair, whereas the analytical bounds describe the exact feasible price response.

Table 23: Complete matched-accuracy deployment on 32 contexts, with three repetitions. All final normalized exact gaps are at most 10^{-7} .

Method	Total (ms)	Audit (ms)	Iterations	Refined (%)
Cold scalar Newton	5.353	5.222	3.75	0.00
Brent	5.363	5.224	7.81	0.00
Sorted breakpoints	5.513	5.222	57.91	0.00
Value only	10.635	10.436	1.69	81.25
Value gradient	10.600	10.397	1.72	78.12

Times are medians including both audits when refinement occurs. Iterations are mean full-objective solver iterations, counting zero for an immediately certified learned response. The classical methods solve directly, so their refinement indicator is zero rather than a claim of zero computation. Breakpoint scan iterations count scanned intervals and are not comparable arithmetic units to Newton or Brent steps.

Table 24: Structural calculation only: median milliseconds over five repetitions at fixed two-date depth. Rational deployment audit is excluded from this scaling table.

Leaves	Threshold	Scalar Newton	Brent	Breakpoints
31	0.007	0.069	0.096	0.096
127	0.006	0.090	0.102	0.256
511	0.008	0.090	0.113	0.893
2,047	0.010	0.131	0.154	3.551
8,191	0.020	0.404	0.360	15.672
32,767	0.067	1.870	1.144	69.810